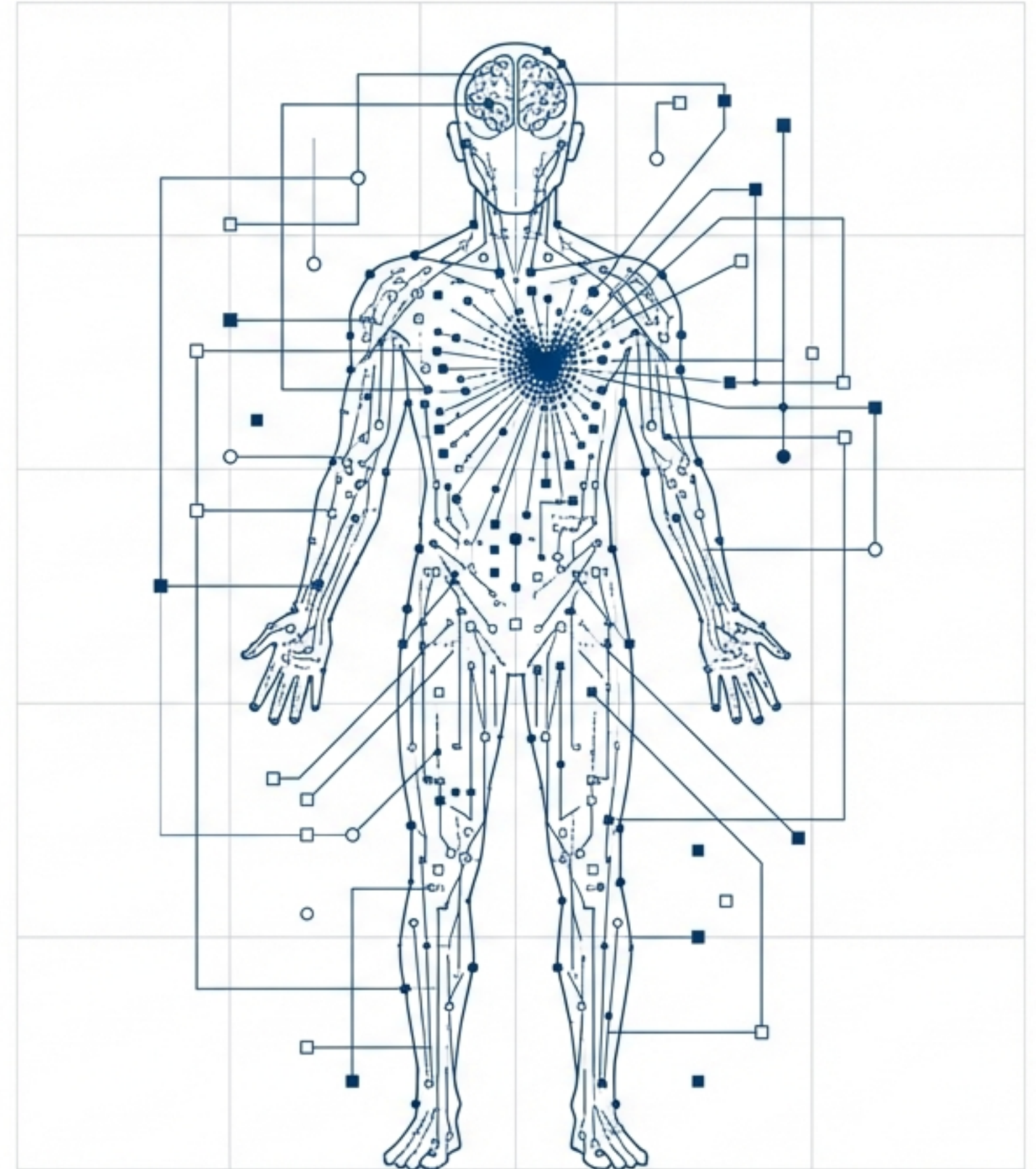


Script Defense Architecture: From Code to Field Action

End-to-End Implementation of the Vi Engage HCP Pipeline (Task 3)

USE CASE: Script Defense (Anti-Churn)

FRAMEWORK: Listen → Think → Act



The Objective: Distinguishing 'Lost Causes' from 'Persuadables'

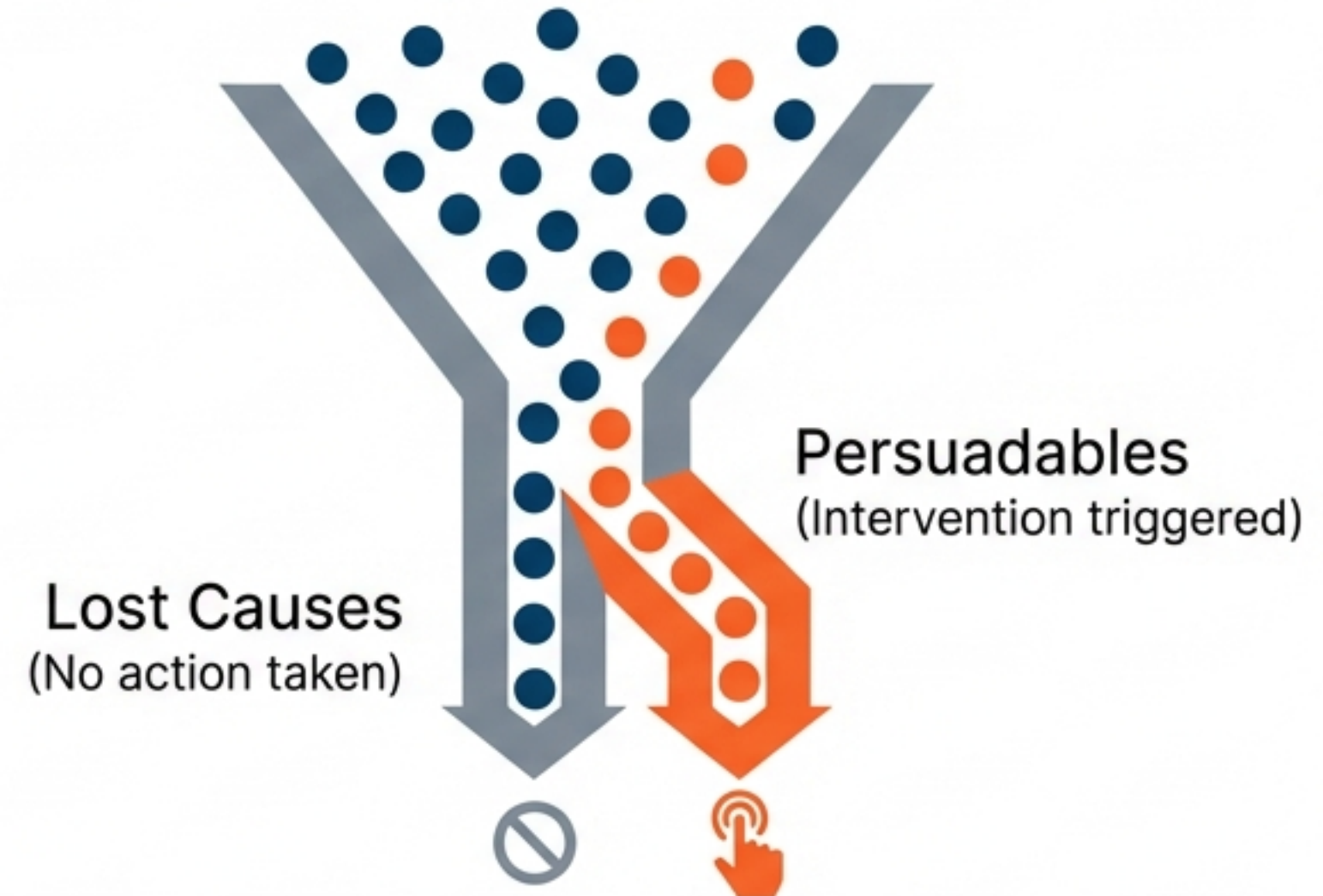
Traditional CRM Approach



Focus: Who will leave? (**Churn Prediction**)

Result: **Wasted resources** on HCPs who would leave regardless.

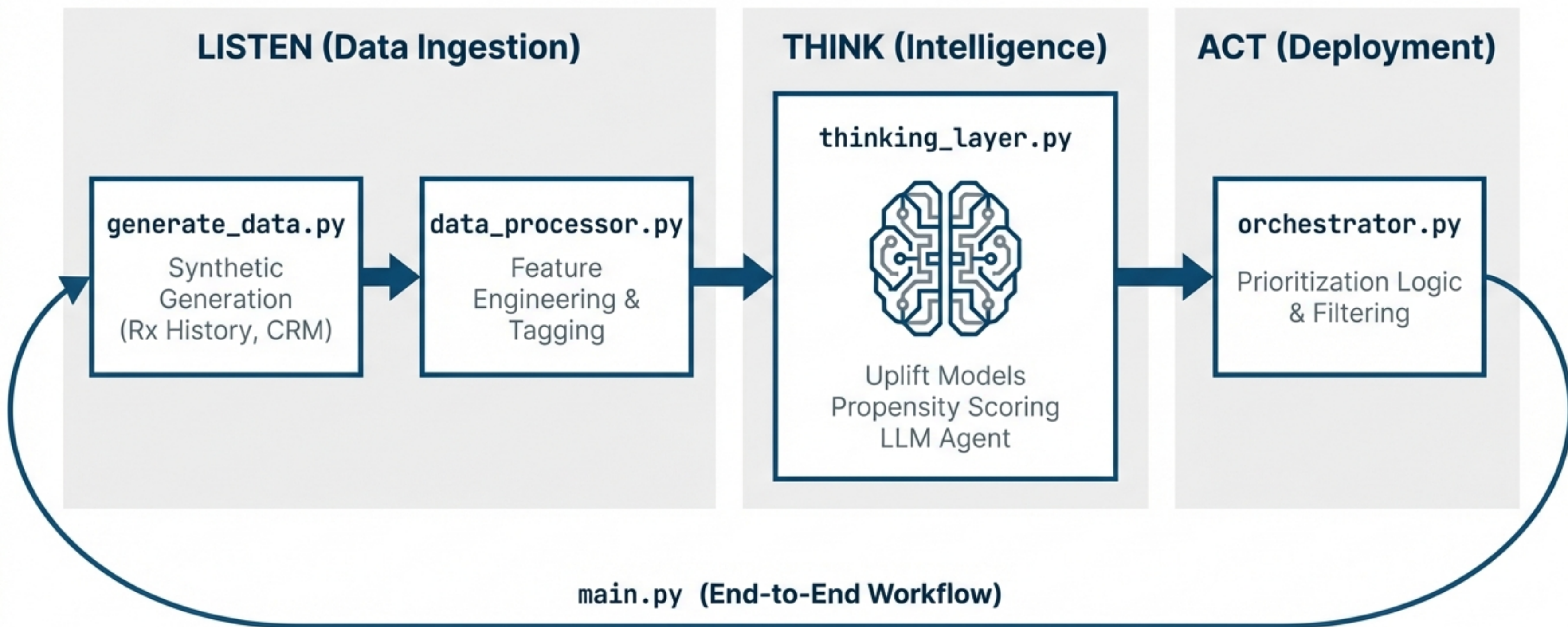
Vi Engage Logic (Task 3)



Focus: Who can we save? (**Uplift Modeling**)

Result: **Prioritized 'Top 10' list** for field intervention.

The Architectural Map

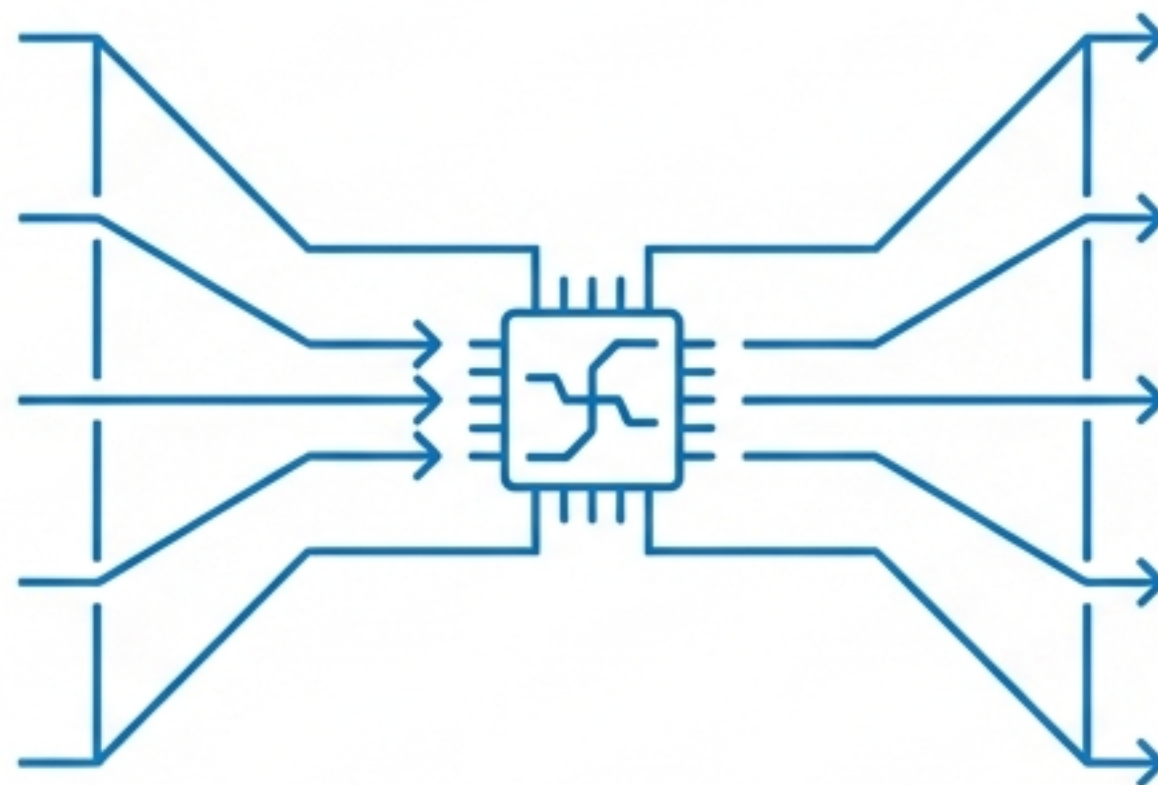


Phase 1: Listen – Synthesizing and Processing Signals

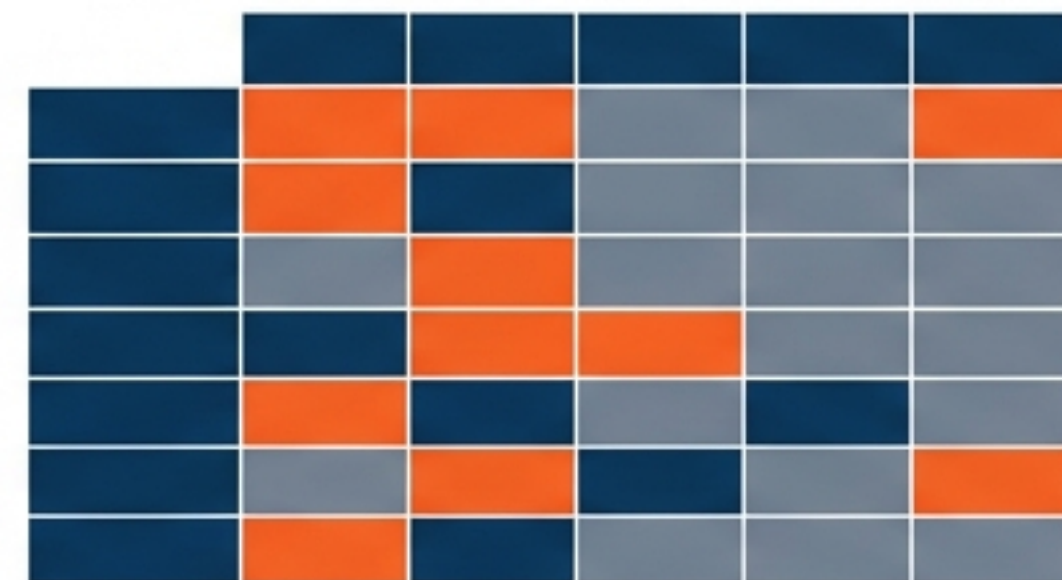
generate_data.py



- 3,000 Synthetic HCPs
- 6-month Rx History
- Field Interventions



data_processor.py



- Feature: current_market_share
- Feature: Rx_trend
- Feature: call_count
- Tagging: Interacted / Retained / Churned

Phase 2: Think – The Dual-Model Approach

File: thinking_layer.py

Track A: Uplift Modeling (Value)

$$P(\text{Retained} \mid \text{Visit}) - P(\text{Retained} \mid \text{No Visit}) = \text{Score} = \text{Intervention Benefit}$$

Identifies Persuadables

Track B: Windowed Propensity (Urgency)

T+1

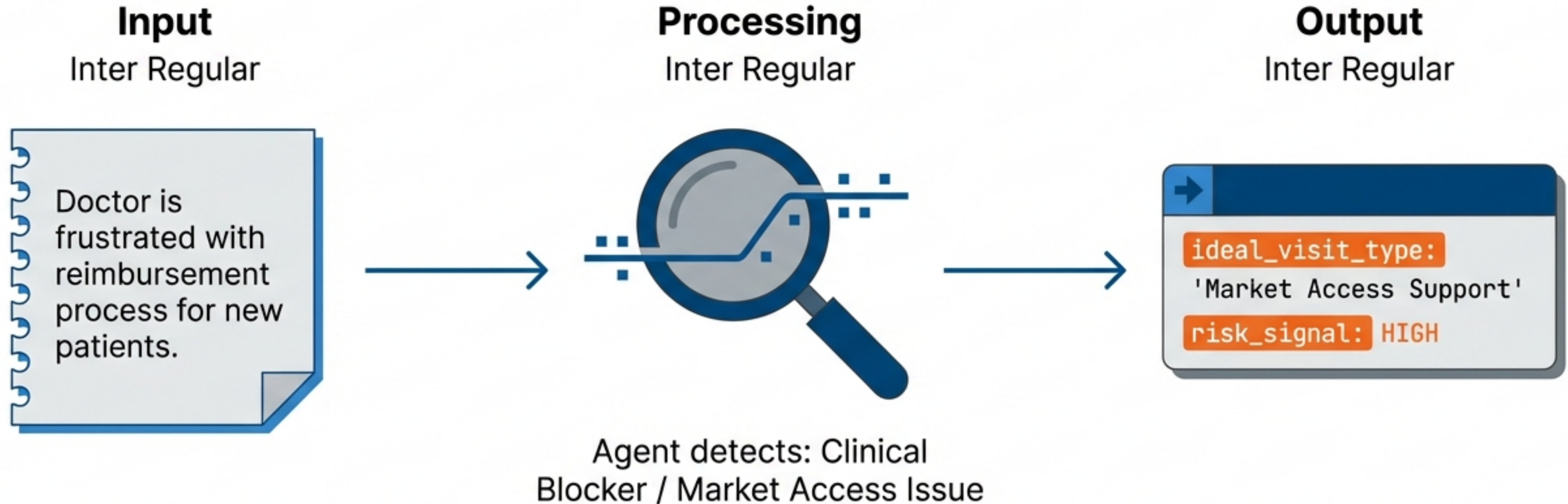
T+2

Identifies Imminent Risk

Composite
Action Score

The GenAI Edge: Parsing Unstructured Field Intelligence

Component: LLM Agent (thinking_layer.py)



Business Value: Translates qualitative notes into quantitative risk signals.

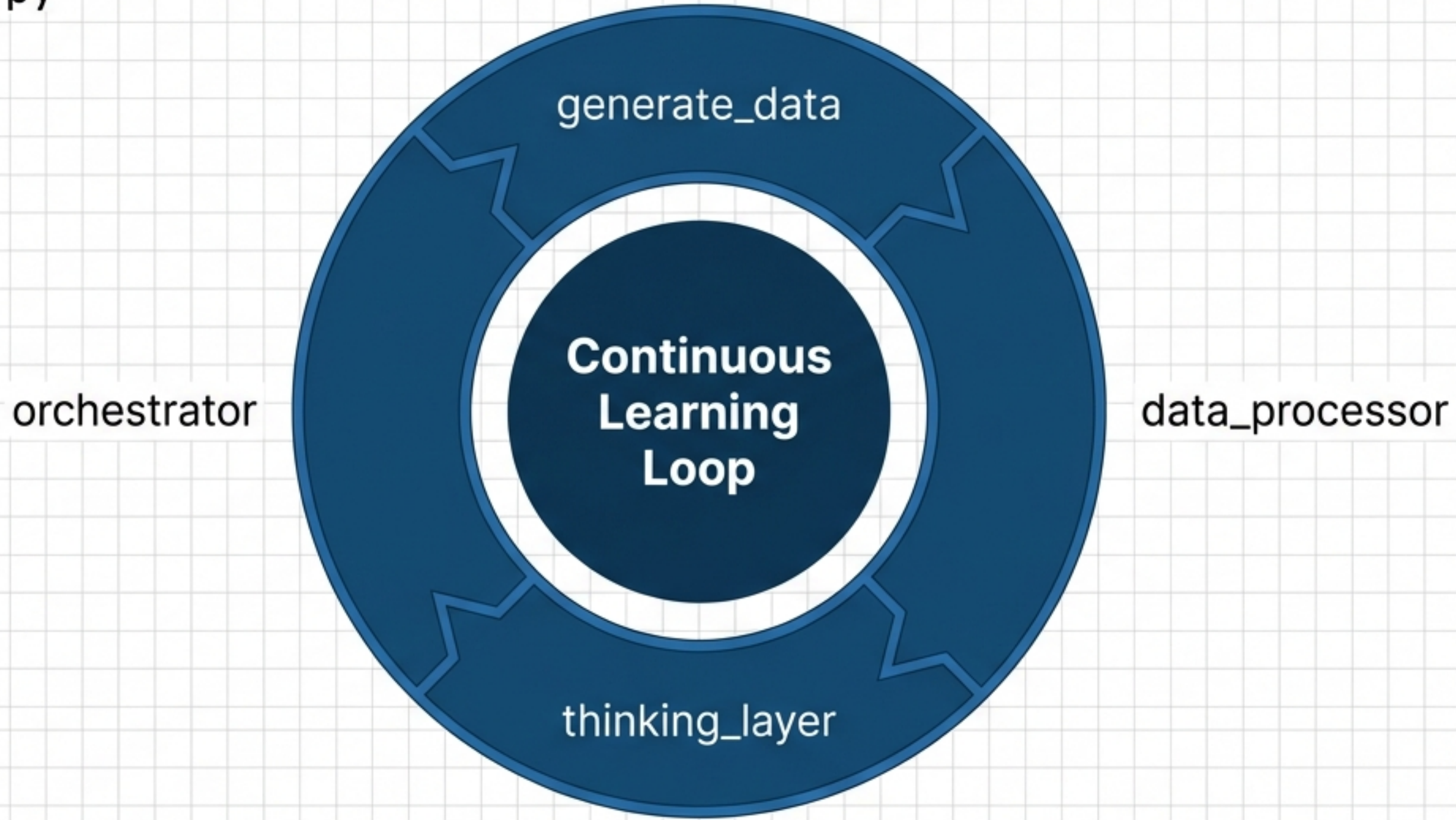
Phase 3: Act – Orchestrating the Next Best Action

File: orchestrator.py



Closing the Loop: End-to-End Execution

File: main.py



Modular design allows for scalable model swapping and continuous retraining.

Impact: Precision Over Volume

File: impact_kpi.py

SAVED SCRIPTS

14%



Pilot vs Control

- ✓ 1. Architecture: Modular Listen-Think-Act loop verified.
- ✓ 2. Methodology: Uplift modeling eliminates waste on 'Sure Things'.
- ✓ 3. Result: Proactive Script Defense replaced reactive fire-fighting.

Vi Engage HCP Pilot Result